Usable Bundle Size

This experiment determines the usable bundle size for given hyperparameters {N,P}.

Test setup:

- Prefill auto-associative memory with 200,000 items.
- For a given bundle size b, randomly select b items from the prefill data.
- Query the memory with the stochastically decimated bundle until an item from the bundle emerges.
- Repeat and compute the median number of queries, needed to extract the first item.

We define the usable bundle size as the number b for which the median number of query attempts is 1 or 2.

The result depends on the decimation factor.

Requirements

```
<< "Circuits.m"
```

Hyperparameters

```
7
```

Test Setup

```
SeedRandom[1];
```

```
RandomSet[n_,p_] := Sort[RandomSample[Range[n], p]];
```

```
extraction[ mem_, mult_Integer, p_Integer] := Module[ {shots = 0, y,
    maxiter = 5 p, sample, bundle, decimate},

    decimate[s_List] :=
        If[ Length[s] > p, Union[RandomSample[s, Round[ decimation * Length[s]

    sample = RandomSample[ data, mult];

    bundle = Union[Flatten[sample]];

    Do[ shots++;
        y = mem["retrieve"][ DDD = decimate[bundle]];
        If[y =!= {} && MemberQ[sample, y], Break[]],
            {i, 1, maxiter}];

    shots
    ];
```

Experiment

```
M = Memory[ <| "A_parameters"→{n, p}, "B_parameters"→{n, p}|> ]
```

```
⟨|A_parameters → {2000, 10}, B_parameters → {2000, 10}, T → 7,
 residual → 2000, store → store$1850, retrieve → retrieve$1850,
 clear :→ clear$1850, memorycount :→ memorycount$1850, backend → C_FFI|⟩
```

```
prefill = 5000000;
data = Table[RandomSet[n,p], prefill];
M["store"] /@ data; // AbsoluteTiming
```

```
{374.628, Null}
```

```
M["memorycount"] / ( n^2 (n-1) /2.)
```

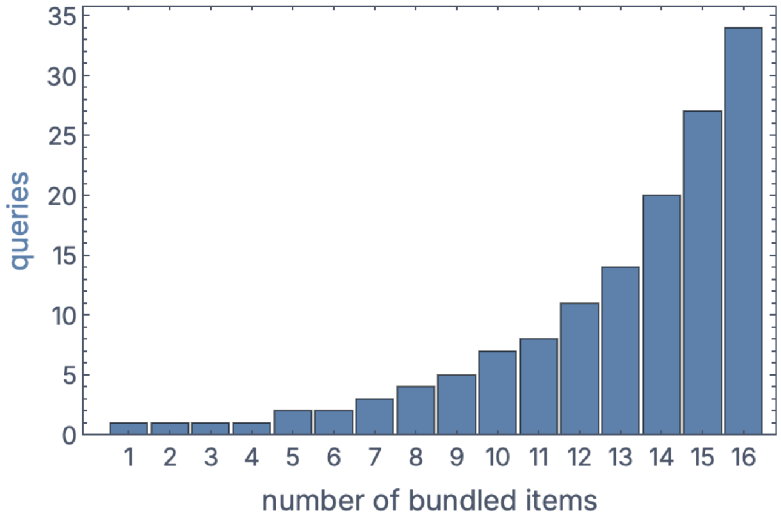
```
0.36343
```

```
testsize = 1000;
maxbundle = 16;
```

```
tab = Table[
    queries = N[Median[Table[ extraction[ M, i, p], testsize]]];
    Print[i, " | ", queries];
    queries, {i, 1, maxbundle}];
```

```
1 | 1.
2 | 1.
3 | 1.
4 | 1.
5 | 2.
6 | 2.
7 | 3.
8 | 4.
9 | 5.
10 | 7.
11 | 8.
12 | 11.
13 | 14.
14 | 20.
15 | 27.
16 | 34.
```

Barchart


queries
35
30
25
20
15
10
5
0
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
number of bundled items